From Embodied Experience to Reusable Relational Knowledge: An Auditable Unified-Graph Architecture and a Preregistered Boundary Analysis of Activation-Based Access

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Abstract

Embodied agents accumulate experience faster than they consolidate it: action histories remain logs instead of becoming reusable knowledge. We present FASCINATOR (FAS), a cognitive-architecture prototype that keeps semantic, episodic, and procedural content on a single typed, weighted graph and writes experience back as relational structure through one evidence-weighted attribution pipeline. The paper contributes an auditable memory architecture together with a preregistered evaluation protocol that delimits when spreading activation—the architecture's access mechanism—adds value over flat retrieval. In an information-asymmetric beacon task, persistent graph context yields 20/20 task completions versus 0/20 without context (McNemar p=2×10^{-6}); an activation-free embedding-retrieval control reproduces the full benefit (20/20). Persistent relational reuse is therefore supported, while the necessity of activation is not. A preregistered mechanism campaign (4×50 paired seeds) rejects activation's incremental value over a closed-form flat baseline (ΔJCG<0, Holm p=1.5×10^{-9}, 0/200 reversals), isolating one exception: multi-input convergence through a shared hub (FAS 30/30 *vs* flat 0/30). An 84-cell parameter sweep maps this exception—FAS strictly beats flat in 11 cells (weak edge weights, shallow paths), loses in 9 (strong weights, deeper paths), and ties in 64. A second, baseline-controlled campaign (360 runs) measures the trade-off directly: under neutral instructions the persistent state resumes an abandoned goal in 9/10 runs where full-history LLM baselines do so in 0/10, and the same persistence costs ten extra failed attempts when the environment changes. All negative results, two assembly audits, and the full protocol are reported.

1 Introduction

As soon as an embodied agent acts for more than a few minutes, it produces a stream of experience: actions, observations, successes, failures. What happens to that stream separates two classes of system. In one class, experience accumulates as logs—available for reflection prompts or retrieval augmentation, but never re-organized into structure that changes future behavior. In the other, experience is consolidated: concrete episodes give way to general relational knowledge that persists, is re-activated in relevant situations, and measurably participates in later decisions. Whether or not the system's core includes a large language model (LLM), the gap between logs and consolidation is a central engineering problem for long-lived agents.

Existing positions are familiar. Cognitive architectures such as ACT-R [1] and Soar [7] consolidate experience through specific mechanisms (chunks, reinforcement) but keep memory stores representationally separate. LLM-based agents [9, 12, 14] typically save experience as text or vectors and reuse it through prompting; consolidation is delegated to the model and is not auditable. A graph offers a middle path: if episodic, semantic, and procedural content share one explicitly typed graph, consolidation becomes a first-class, auditable operation rather than a side effect of prompting.

This paper reports FASCINATOR (FAS), a Python prototype built around that idea: a *unified cognitive graph* on which spreading activation [3] drives attention and candidate generation, an attribution pipeline writes experience back as typed relational structure, and ledger and event-framework channels route stored knowledge into behavior. We deliberately keep two propositions separate throughout:

(a) **Persistent relational reuse has behavioral value.** Under information asymmetry—the knowledge needed for a decision exists only in memory, not in the current observation—graph context built

from experience changes behavior (beacon task: 20/20 completions vs 0/20 without context).

(b) **The necessity of activation is not supported.** The same benefit is fully reproduced by activation-free retrieval (D-flat, 20/20), and a preregistered mechanism campaign rejects activation's incremental value over flat retrieval under all four confirmatory conditions.

The contributions are:

1. **An auditable unified-graph memory architecture.** Semantic, episodic, and procedural content coexist in one typed weighted graph; experience becomes new relational structure only through a single evidence-weighted write-back channel. Isolation experiments verify the channel (0 vs 13 learned nodes under an identical experience stream), cross-context re-activation (4/12 vs 0/8 vs 0/0 marked nodes), and two consumer channels that measurably flip action selection (Sections 3–5.2).

2. **A preregistered evaluation protocol with honestly reported negative results.** The campaign testing whether spreading activation adds value over flat retrieval was designed, seeded, and committed before execution; its primary hypothesis was rejected (ΔJCG<0 under all four preregistered conditions, Holm p=1.5×10^{-9}, 0/200 reversals), and we report this as a primary result (Section 5.3).

3. **A boundary analysis of activation-based access.** The single systematic exception—multi-input convergence through a shared hub (30/30 vs 0/30)—is delimited by an 84-cell parameter sweep over edge weight, input count, and path depth, and complemented by a structural diagnosis of relation-direction semantics (Section 5.3, 6).

Every number in this paper is programmatically extracted from archived run outputs; the evidentiary standard, the deterministic-sandbox protocol, and the statistics are specified in Section 4. Internal codenames, per-family provenance, and full per-experiment data are in the Supplementary Material.

2 Related Work

Cognitive architectures. ACT-R [1] and Soar [7] treat consolidation as mechanism-specific learning over separate memory stores; OpenCog [5] shares our atom-space motivation but targets general intelligence rather than auditable consolidation of embodied experience. Episodic-memory theory [2, 13] and event segmentation [11, 15] motivate our event-framework channel and attribution window. Spreading activation as an access mechanism goes back to Collins and Loftus [3]; attention and working-set selection relate to Cowan [4], Itti and Koch [6], Posner [10].

LLM-agent memory. ReAct [14], Generative Agents [9], Reflexion [12], and RAG [8] reuse experience through prompting or vector retrieval; consolidation quality is not separately controllable or auditable. FAS uses the LLM only as a budgeted interface for language and never as memory (Section 3.5); the studied knowledge channels run with the LLM hard-disabled.

Relation to our contribution. None of these lines reports a preregistered test of whether activation-based access beats flat retrieval on the same structure. Our negative result on that question, and the parameter sweep delimiting the exception, are the paper's methodological core.

3 The Fascinator Architecture

We present the system content-first: the graph, activation, write-back, persistence and consumers, then the LLM boundary.

3.1 The unified cognitive graph

FAS maintains a directed, typed, weighted graph $G = (V, E, W)$. Nodes carry a label τ_v marking memory role (seven classes—items, recipes, action experiences, changes, events, intentions, gaps—sharing one namespace), a base weight w_v, a timestamp, and a runtime activation $a_v \in \mathbb{R}$; each edge $e = (u, v, \rho, w)$ carries a relation type ρ (eight classes: produces, belongs-to, drops, needs, needs-tool, leads-to, involves, results) and a static weight $w \in [-2, 2]$; negative weights denote suppression. *There is no separate episodic, skill, or intention store*: goals, events, action patterns, and affect are nodes of the same graph, distinguished only by label and relation type (Fig. 1). Beside the graph there are only two auxiliary layers—a short-term episodic buffer (capacity 100, for short-horizon integration) and a statistical timeline (the ledger)—which

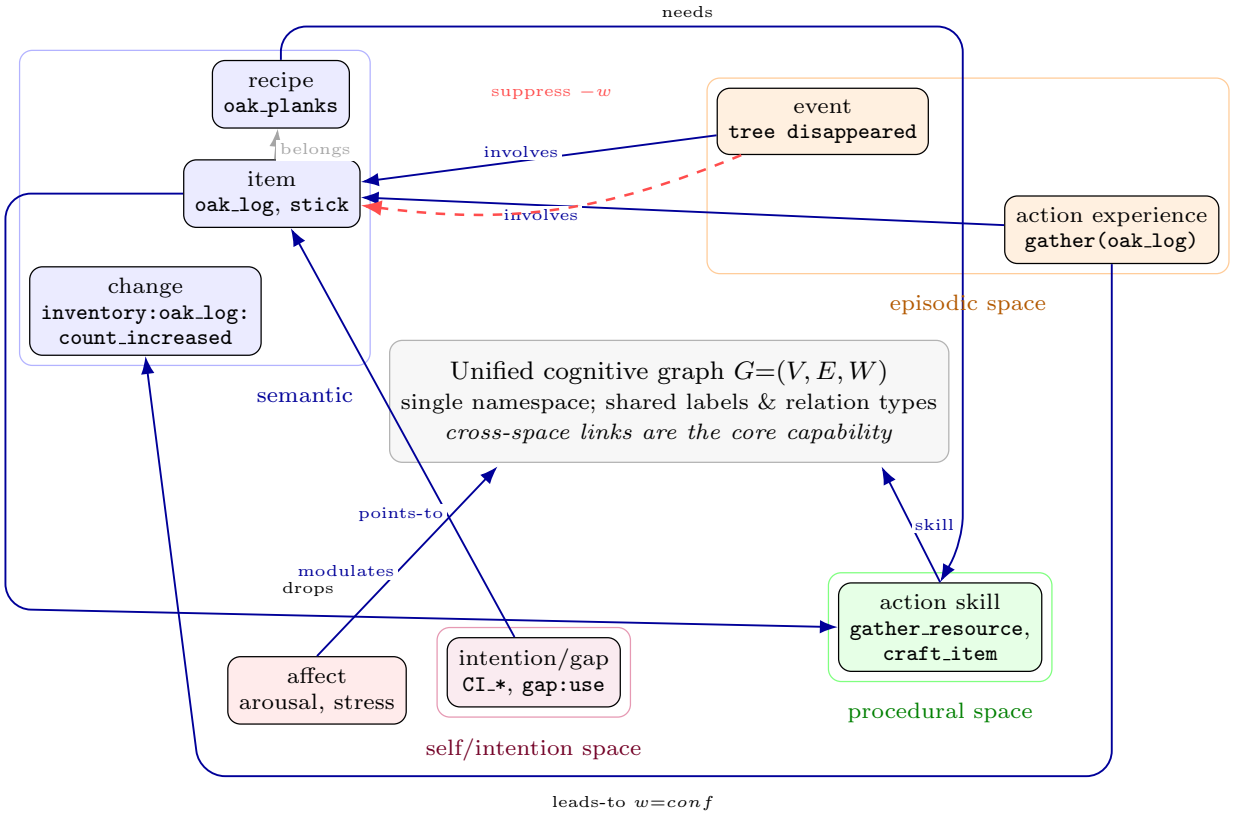


Figure 1: The unified graph ontology (schematic). Seven label classes and eight relation types share one namespace; there is no separate episodic, skill, or intention store. Cross-space links (blue, corridor-routed) are the core capability: the `leads-to` edge from `action experience` to `change` (weight = confidence) is the product of the write-back pipeline (Section 3.3); `gap` points to an entity to inject activation; `event` carries a negative-polarity slot edge (red dashed) suppressing the entity (Section 3.4).

never build structure: the sole structure-writing channel is the attribution pipeline of Section 3.3. The space split serves lifecycle management only; it does not isolate topology.

3.2 Cognitive state and spreading activation

The cognitive state at time t is the activation field $\mathbf{a}(t)$. Inputs (observations, intentions, gap injections) deposit initial energy into the corresponding nodes; each step, every active node u emits to its neighbors

$$\Delta a_{u\to v}(t) \;=\; a_u(t)\,\cdot\,\tilde{w}_{uv}\,\cdot\,\beta_{\text{cat}}(\rho)\,\cdot\,g_{\text{rel}}(\rho)\,\Big/\,\textstyle\sum_{v'}\max\big(\tilde{w}_{uv'},0\big), \tag{1}$$

followed by decay $a_v(t{+}1) = (1-\lambda)\,a_v(t) + \sum_u \Delta a_{u\to v}(t)$, clamped by per-space maxima (episodic space $a_{\max}$=3, global default 5). Emission follows an "emit = transfer" policy: each node's emitted quantity per step (default 1.0) is deducted from itself; source nodes are exempt. This is an engineering choice of short memory, not an energy-conservation law. Negative-weight edges contribute suppressive flow; suppression is issued only when the emitting node's positive-weight total is greater than zero. The emission budget is scaled by a modulatory gain

$$g \;=\; \text{clip}\big(1 + 0.3\,\text{arousal} - 0.15\,\text{stress},\; 0.7,\; 1.6\big), \tag{2}$$

where arousal is derived from affective-node activation and stress from a slow tonic offset. Attention is the Top-k of the field and constitutes input to action selection. Fig. 2 sketches activation-based access *side by side with its flat-retrieval comparison*; the schematic builds intuition only—all conclusions are carried by the experiments of Section 5.3.

3.3 Experience attribution and write-back

After an action settles, its receipt and the attribution window (8–120 s of environment time) of observations form candidate action–observation pairs. Repeated pairs aggregate into hypotheses carrying support s,

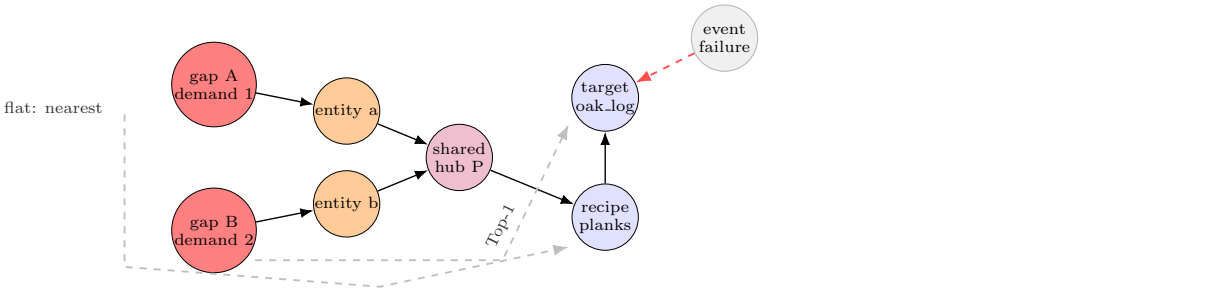


Figure 2: Spreading activation and flat retrieval side by side. Every numbered claim comes from the archived experiments of Section 5.3; the figure only builds intuition.

Algorithm 1 Attribution–write-back: experience to relational knowledge

Require: action–observation stream (a_t, o_t) within the window; ledger $\mathcal{L}$; graph G

For each action A_t, associate observations O_k in the window with its settlement as candidate pairs (A_t, O_k)
Aggregate by action signature: $s \mathrel{+}= 1$ (positive outcome), $c \mathrel{+}= 1$ (contra)
conf $= s/(s + c + 2)$ (Laplace smoothing)
if $s \geq 5$ and conf ≥ 0.80 **then**
    create `action:`A, `change:`O nodes if absent
    write edge `action` $\longrightarrow$ `change` with relation `leads-to`, weight w=conf; mark both active
end if
ledger in parallel maintains per-action success priors $\mathcal{L}$.prior(sig(A)) consumed by action selection

contra-evidence c, and Laplace confidence conf $= s/(s+c+2)$. When $s \geq 5$ and conf ≥ 0.80, the hypothesis is *promoted* to a typed relation on the graph: a `leads-to` edge of weight $w =$ conf from the action node `action` :$A(x)$ to the change node `change` :O,

$$\texttt{action}:A(x) \longrightarrow \texttt{change}:O \qquad (\texttt{leads-to},\ w{=}\text{conf}), \tag{3}$$

and both endpoints are mark-activated. This is how experience becomes *new relational structure*: `action:`/`change:` nodes exist only as products of this pipeline. The same ledger maintains per-action success-rate memories, consumed by action selection as prior bonuses (±0.05)—the second statistical path from experience to behavior. A third channel encodes discrete events as event-framework nodes with polarity-slot edges (implemented and production-default off; this paper assembles it explicitly, Section 5.2). The algorithm is Algorithm 1.

3.4 Persistence and consumers

Knowledge survives episode boundaries by construction (the graph is inherited, not reinitialized); persistence carries two payloads: *structure* and *activation values* are both serialized into the next episode, so "re-activation across contexts" is a measurable event. Section 5.2 measures that re-activated ($a \geq 0.05$) differences are carried exactly by episodic structure connecting shared entities (4/12 vs 0/8 vs 0/0). Snapshot rotation and a shrink guard (save refusal to overwrite a running graph with an empty one) are data-integrity mechanisms, not cognitive content.

Four classes of consumer use the knowledge produced by write-back. **(a) Ledger channel (production):** candidate action scores are reweighted by ledger success priors (+0.05 success / −0.05 failure). **(b) Affect/neuromodulation (production, every tick):** arousal and stress scalars modulate the emission gain via Eq. 2. **(c) Continuous intentions and replay gating (production):** the continuous-cognition cycle runs graph dynamics every 2.5 s; intention lifecycle and idle-gated offline experience replay belong here. **(d) Event framework (experimental assembly, production-default off):** negative events suppress entity activation via polarity-slot edges; the code is production-hardened but disabled by default so that nothing lands on the graph in production; all behavioral evidence comes from experimental assem-

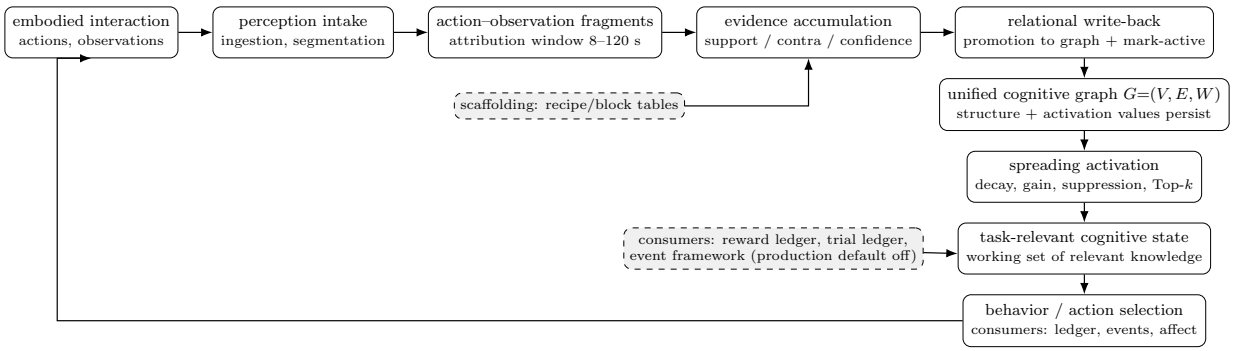


Figure 3: The FAS conceptual loop: from embodied interaction to reusable relational knowledge, and back to behavior via activation and consumers. Dashed blocks are scaffolding (mechanics tables; the necessary precondition characterized by the zero-prior result, Section 5.6) and consumption channels. The LLM's only positions in the loop are language production/interpretation/reflection (Section 3.5).

bly. *Experimental layers* (sandbox assembly and the LLM decision head of Section 4.4) never enter the production default path; the run recorder only observes—it never modifies cognitive behavior.

3.5 The LLM as a budgeted interface

FAS uses the LLM for language production, interpretation of unstructured input, and (low-frequency) reflection; it never uses the LLM as memory, nor as the source of the knowledge this paper studies. All subject experiments run under an import-time guard that hard-disables the LLM (zero-LLM contract; any call errors out)—the only exceptions are the explicitly labeled decision-head experiments (Section 4.4).

3.6 The experience-to-knowledge loop

Fig. 3 shows the complete conceptual loop. Experience is segmented into temporally ordered action–observation fragments (the attribution window), repeated patterns accumulate evidence into candidate relations, and, after support and confidence checks, are written back as persistent graph structure. New inputs unfold across this graph via spreading activation, forming a task-relevant cognitive state with a Top-k focus; consumers carry that state into action selection; actions change the environment, producing new experience. Language production, interpretation, and reflection are the only LLM calls in the loop (Section 3.5): the LLM is a consumer/decision component, not memory.

4 Evaluation Protocol

4.1 Environment, determinism, and what "paired seeds" mean

All experiments run in a deterministic `SandboxWorld`: a discrete Minecraft-style grid world with a simulated clock (16 s/tick), recipe and block tables, and a fake device bridge replacing a real Minecraft LAN bridge. The agent stack is the production stack—knowledge graph, diffusion engine, continuous-cognition components, autonomy loop, action manager, and the real skill layer—with the LLM disabled by the import-time guard. One long-chain condition uses a documented simplification (sandbox smelting settles instantly into the inventory); results relying on it are marked.

Every condition is instantiated with a fixed seed; for the zero-LLM subject experiments, identical seeds reproduce bit-identical trajectories (verified by re-runs). For the LLM decision-head families, see the determinism re-run in Section 5.4. *Paired seeds* instantiate different random instances of the same condition and support paired statistics across instances; because the world is a deterministic sandbox with synthetic topologies, they do *not* sample real-world variance. Statistical tests in this paper therefore quantify the consistency of a direction of difference across instances, not population-level effect sizes. Sample sizes: subject experiments n=5 seeds; mechanism campaign n=50 paired seeds per condition; beacon and routing n=20 paired seeds; relation-direction diagnosis n=10 paired seeds. Every run is archived by the run recorder (metadata, event stream, start/end graph snapshots, metrics) into content-addressed directories; the archival census is given in the Supplementary Material (Table S2).

Table 1: Evidence levels × protocol layers (merged scale; used to attribute every claim in this paper).

	Level of the claim	**Layer the evidence comes from**
L1 representational	learned structure exists and re-activates	production default path (zero-LLM subject behavior)
L2 mechanism	measurable activation/ledger dynamics	production mechanism under experimental assembly; mechanism-isolation layer (reference models distilled from production semantics)
L3 behavioral	stored knowledge changes action selection	assembled consumer channels; decision-head harness (experimental layer, never attached to production)
L4 general capability	—	*explicitly not claimed anywhere in this paper*

4.2 Preregistration and evidence accounting

The mechanism campaign's hypotheses, topologies, seed reservoir, metrics, and testing scheme were fixed and committed before execution; deviations (one assembly repair before unblinding) are reported as such. Evidence is reported at three levels—*representational* (existence and re-activation of learned nodes/edges), *mechanism* (activation trajectories, weight changes, ledger counts), and *behavioral* (completion ticks, action counts, first-action identity)—and attributed to one of four protocol layers (production default; production mechanism under assembled validation; mechanism-isolation layer on reference models; experimental-layer harness). Table 1 merges the two scales. General-capability claims ("L4") are explicitly not made. Where the word "causal" is used, it means rule-like regularities inferred from temporally ordered action–outcome observations with evidence weights—not interventional identification; no do-style test is performed anywhere.

4.3 The mechanism-campaign protocol (preregistered)

The core incremental-value campaign studies the *incremental* value of spreading activation over flat retrieval. Four synthetic subject topologies are used (3 inputs/100 nodes/25% density; 3 inputs/300 nodes/50%; 5 inputs/300 nodes/50%; 5 inputs/1000 nodes/75%). Four candidate generators are compared: **B1** direct injection (graph-free); **B2** *flat path-product*—a closed-form approximation of retrieval on the same structure (per input–candidate pair, the best simple-path edge-weight product, summed over inputs; same structural information, no iterative propagation; note this is a closed-form surrogate, not an embedding retriever—the beacon task's flat control in Section 5.1 is a genuine sentence-embedding retriever); **B3** random activation (equal-energy control); **B4** oracle (ground truth, used only for generator-correctness prechecks; MRR=1.0 on all four topologies). The primary metric JCG (joint convergence gain) measures multi-input convergence onto shared targets. The preregistered confirmatory conditions C-1–C-4 (n=50 paired seeds each) use one-sided paired Wilcoxon with Holm correction to test H3 (*diffusion's JCG exceeds flat's*). Ancillary families (all run and archived): insertion-order permutations, label/ID permutation (hidden oracle), a no-joint-target counterfactual, a negative-edge test, a weight sweep $w \in \{0.1..0.9\}$, noise/distractor sweeps, and combinational-topology detectors. Condition-level numbers are in the Supplementary Material (Table S6); the preregistration ships with the artifacts.

4.4 LLM decision-head protocol (routing experiments)

One family places a real LLM at the decision point (single model, temperature 0, one call per decision). Menu, environment state, task facts, experience stream, and per-decision budgets are identical across conditions; the only difference is the context the model sees: (A) none; (B) raw last-8 experiences; (C) embedding-similarity retrieval top-5; (D) FAS mechanism-selected context (graph entry, propagation, Top-k focus, plus cognitive demand/gap/routing annotations); (E) random memory entries (counterfactual control). Four task classes are used, all fully observable at decision time. The repaired official campaign runs on the production edge-building path with nine fail-fast harness invariants (520/520 runs valid). The campaign is positioned as exploratory: it measures context-selection value, and the FAS loop does not control execution.

Table 2: Beacon task results (n=20 paired seeds per condition; 80 runs). All numbers from the archived campaign analysis.

Condition	Completed	Decisions (mean, median)	First action `gather_oak_log`	Contrast
fas_full	20/20	2.15, 2	20/20	vs direct: McNemar $p=2\times10^{-6}$
D-flat (embedding retrieval)	20/20	2.1, 2	19/20	vs fas: p=1.0; decisions Wilcoxon p=0.72
D-noact	0/20	8.0, 8	20/20	vs fas: McNemar $p=2\times10^{-6}$
llm_direct	0/20	8.0, 8	17/20	—

4.5 Reproducibility

Fixed seeds; single-process deterministic execution; no wall-clock-dependent logic in the measurement path (one documented exception above); archived raw runs and content summaries; 104 regression tests pass at the time of writing. A determinism re-run check for the LLM decision-head family is reported in Section 5.4.

5 Results

5.1 Persistent relational reuse: the information-asymmetric beacon task

Task. The beacon task operationalizes information asymmetry: the knowledge needed to solve the task exists *only* in experience-written memory, in no input prompt. The target recipe is stripped from all task facts ("closed book"); the crafting menu lists `craft_X` only while item X's context is present (menu gating); goal entities are excluded from activation seeding. Four conditions, n=20 paired seeds, temperature 0, all 80 runs valid: **fas_full** (persistent graph + activation context), **D-noact** (same graph context with activation dynamics disabled), **D-flat** (flat retrieval without activation aggregation), **llm_direct** (no memory context).

Results. fas_full completes 20/20; llm_direct completes 0/20 (McNemar exact $p=2\times10^{-6}$; every decision window is used up without unlocking the chain). **D-noact**—structure present, runtime access absent—fails 0/20: structure alone is not usable access. **D-flat**—activation-free flat aggregation—fully reproduces the benefit (20/20; decisions mean 2.1 vs 2.15, no statistical difference, Wilcoxon p=0.72). First-action identity gives a mechanism fingerprint: fas and D-noact both choose `gather_oak_log` 20/20, direct only 17/20—the difference is not in first-action choice but in subsequent unlocking.

Two readings must be kept separate. **(a) Persistent relational reuse produces behavioral benefit** on a high-noise, recipe-opaque, information-asymmetric task. The 0/20 baselines are, by task design, an *existence verification*: closed-book gating makes the recipe unavailable to context-free conditions by construction, so the fas-vs-direct contrast demonstrates that the pipeline delivers the required knowledge through memory, not that memory context is "statistically significantly better" in any population sense. **(b) The necessity of activation is not supported**: D-flat's complete reproduction joins the mechanism campaign of Section 5.3—flat retrieval suffices on reachable tasks.

What kind of retriever is D-flat? D-flat is a genuine sentence-embedding retriever: it encodes all graph node identifiers with a public, locally runnable sentence-transformer model (BAAI/bge-small-zh-v1.5) and ranks them by cosine similarity to the task text—no activation, no propagation, no path structure. A zero-LLM accessibility check (Supplementary Material S6) verifies the enabling condition directly: for the beacon goal text, the recipe node `item:oak_planks` ranks 2nd and the resource node `item:oak_log` 4th among all 44 graph nodes, inside the Top-8 context window, for all 20 seed assemblies. A complete real case study of the wood-to-plank chain (practice experience, write-back, persistent graph, beacon behavior) is shown in Supplementary Figure S3.

5.2 Write-back isolation, persistence, and consumption

Write-back isolation. If written-back structure were a rearrangement of injected knowledge, disabling write-back would change nothing. The isolation experiment breaks graph write-back on a single switch:

perception, actions, and environment are identical; two lineages practice the same four-step chain for eight rounds in the same world. With write-back off, the final graph contains **0** learned `action:/change:` nodes; with write-back on, **13** (Supplementary Figure S1a). Sorting the 15 promoted relations of the write-back-on lineage (Supplementary Table S4): 11 substantive action–effect associations ("mining yields items and removes a block"), 4 degenerate relations (action pointing at its own success event—the promotion rules do not distinguish this class; we report rather than hide it), and 2 substantive pairs polluted by the attribution window (downstream craft effects attributed to the upstream gather). The initial recipe tables know *what can craft what*; *what execution produces* can only be established by doing.

Cross-context re-activation. Episode 1 practices the chain in one context; Episode 2 changes the surface context completely (interfering stand, different layout) with a structurally similar goal. Three conditions inherit from Episode 1: full graph (G0), flat-memory control with episodic-space nodes stripped (G1), and a clean graph (G2). Across 5/5 seeds: G0 re-activates **4 of 12** marked nodes ($a \geq 0.05$); G1 **0 of 8**; G2 has none (Supplementary Figure S2). The re-activation difference is carried exactly by the episodic event structure connecting shared entities; behavioral metrics are identical across conditions—representational/access-layer differences are separated from the behavior layer.

Ledger consumption. After 8 practice rounds of a three-step chain (ledger promotes 7 relations), the agent migrates to a structurally related four-step task with the ledger inherited (Transfer) or fresh (No-Transfer). Deterministic, 5/5 seeds: Transfer completes in 13 ticks / 8 actions; No-Transfer needs 16 ticks / 9 actions; the first action flips from a curiosity-driven detour to the experienced resource, and the difference lands exactly on the three candidate actions shared by both tasks (prior (1.0, obs=8) *vs* (None, 0)). Task success is 5/5 in both conditions (Supplementary Figure S4).

Event-framework consumption. A negative event (a tree that disappears on first dig) is encoded as an event-framework node whose negative-polarity slot edge suppresses entity activation. After three warm-up ticks, the suppressed curiosity path loses the first-action competition: 5/5 control first-actions at tick 9 *vs* inhibited action at tick 6 (3 ticks earlier, 1 curious action saved). With *real* failure receipts (written only when an action genuinely fails), control retries the doomed target once (n=4, first completion attempt @17), the event-framework condition never (n=3, @15)—and because that failure mode is a transient timeout, the statistical channel is fully silent on it: the graph-side event framework is the only trace of the failure and the only source of the behavioral difference. Scope: this channel is assembled at the experimental layer and production default is off. It is evidence that graph-encoded experience *can* flip action selection, not evidence that production does so today.

5.3 Preregistered boundary analysis of activation-based access

The convergence mechanism genuinely runs. The mechanism campaign first asks whether multi-input convergence on the graph is a real phenomenon: JCG>0 on **200/200** paired seeds (each confirmatory condition n=50; median JCG 0.0977/0.0977/0.2158/0.2177); the hypothesis group ($\geq$2 inputs) reaches MRR@1=1.0. Counterfactual controls behave as required: with no joint target, all four methods score 0.0 (convergence is not a self-fulfilling prophecy); under label/ID permutation (hidden oracle, 30 seeds), target rank 22.5 is unchanged (no structural privilege); insertion order (10 permutations) gives CV=0.0; the negative-edge test leaves totals unchanged (0.0333=0.0333; total 17.41). An independent 663-run falsification campaign had first confirmed the convergence reference model (JCG>0 rate 0.970). At the access layer, structural diffusion is also distinguishable from noise and from lookup: on a four-condition access task, full diffusion produces activation quality 61.5 over a field of 39 lit nodes, diffusion-off 16.6 over 17, the equal-energy random control 45.8 over a larger structureless field, and a target-chain oracle 17.2 over 18 (Supplementary Figure S5).

Incremental value over flat retrieval: rejected. Against the closed-form flat path-product (B2), all four preregistered conditions show ΔJCG<0 (median differences -0.402, -0.402, -0.784, -0.782; paired n=50 per condition; one-sided Wilcoxon with Holm correction, $p=1.5 \times 10^{-9}$; effect r=0.615; seeds where FAS beats flat: **0/200**). MRR: FAS below B2 (0.0435/0.0426 vs 0.5/0.5 on the two smaller topologies; tie at 1.0 on the two larger). Noise robustness is unsupported: robustness-sweep slope 0.0076 (B2 0.0), AUC 0.0427 vs 0.04. **H3 is rejected**: under these topologies and information budgets, flat retrieval is no worse than diffusion, and usually better (Table 3, Fig. 5a).

The shared-hub exception, parameterized. The only systematic positive difference in the preregistered campaign appears in the combinational topology: multiple inputs' relevant paths converge through a *shared hub* (FAS 30/30; B2 0/30, B1 0/30, B3 5/30; single weight configuration w=0.5). Because this exception carried the paper's only positive evidence for activation, we ran a preregistered-style parameter sweep (design frozen before execution; Supplementary Material S8): joint grid over hub-path edge weight $w \in \{0.3, \ldots, 0.9\}$, number of converging inputs $k \in \{2, 3, 4, 6\}$, and private-path depth $d \in \{0, 1, 2\}$, 30

Table 3: Preregistered boundary analysis: key mechanism-level numbers. All values are read programmatically from the archived statistics files; full data in Supplementary Material S7–S9.

Test	Numbers	Reading
JCG>0 (mechanism runs)	200/200 paired seeds; medians 0.0977/0.0977/0.2158/0.2177	convergence mechanism is real
ΔJCG (C-1–C-4, preregistered)	−0.402/−0.402/−0.784/−0.782; Holm p=1.5 × 10^{-9}, r=0.615, 0/200 reversals	**H3 rejected**
MRR (FAS vs B2, smaller topologies)	0.0435/0.0426 vs 0.5/0.5	flat better on reachable tasks
Noise robustness	slope 0.0076 vs 0.0; AUC 0.0427 vs 0.04	no diffusion advantage
Shared-hub topology (single weight w=0.5)	FAS 30/30 vs B2 0/30, B1 0/30, B3 5/30	the one systematic exception
Weight sweep	both methods reach MRR=1.0 together at w=0.7	structural threshold, not a FAS advantage
Dilution	ρ(reach, act): −0.445 (noise), −0.656 (distractor)	wider spread, flatter target
Shared-hub parameter sweep (84 cells × 30 seeds)	FAS strictly better in 11 cells (all $w\leq0.5$, $d\leq1$); flat strictly better in 9 (all $w\geq0.7$, $d\geq1$); tie in 64; O3 replication 30/30 vs 0/30	advantage region: weak edges × shallow convergence paths

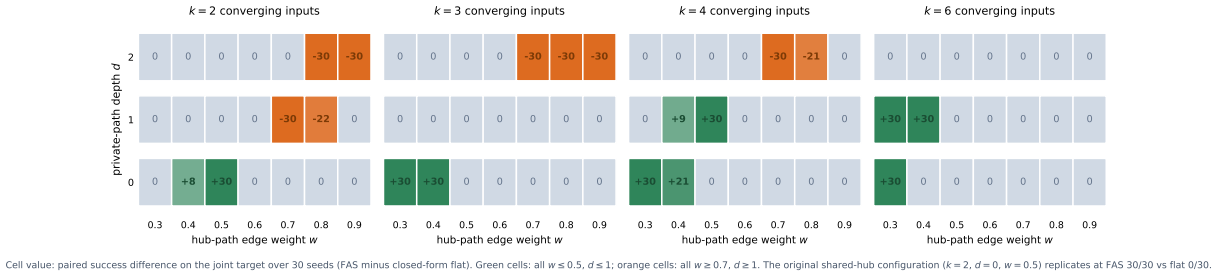


Figure 4: Shared-hub parameter sweep (84 cells × 30 paired seeds, deterministic, zero-LLM). Cell color: paired success difference (FAS minus flat) on the joint target; green = FAS strictly better, orange = flat strictly better, gray = tie; annotations give the cell counts. The FAS-advantage region is confined to weak edge weights and shallow convergence paths. All numbers read programmatically from the archived sweep outputs.

paired seeds per cell, FAS vs the same closed-form flat baseline, with the original single-hub configuration as a hard preflight (replicated: FAS 30/30 vs B2 0/30). The result is a mapped surface, not a point (Fig. 4): FAS is strictly better in 11 of 84 cells—*all* with weak weights ($w\leq0.5$) and shallow convergence paths ($d\leq1$); flat is strictly better in 9 cells—*all* with strong weights ($w\geq0.7$) and deeper paths ($d\geq1$); the remaining 64 tie. On the O3-shaped axis (k=2, d=0), flat catches up exactly when the hub-path weight reaches 0.6 (both methods 30/30 at $w\geq0.6$). The mechanism reading is consistent with the dilution result: convergence helps when the hub can accumulate multi-input energy faster than single paths saturate their targets, and hurts when strong edges let flat ranking resolve the target directly or when long weak paths dilute the spread.

Weight threshold, not diffusion advantage. In the campaign's weight sweep, both methods jump to MRR=1.0 together at w=0.7 (FAS 0.083@0.5→1.0@0.7; B2 0.077→1.0)—a structural resolvability threshold, not a FAS-specific advantage (Fig. 5b). The dilution trade-off completes the picture: the wider the diffusion front (more reachable nodes), the lower the target activation (ρ=−0.445 noise sweep, −0.656 distractor sweep; Fig. 5c).

Relation-direction semantics (diagnostic). A separate mechanism diagnosis (10 paired seeds; full data in Supplementary Material S9) showed that the production recipe graph is a depth-1 star forest whose relation targets are all sinks, making forward two-hop propagation structurally impossible; adding generic reverse-edge projections restores multi-hop reachability and roughly doubles the rank correlation between topological score and activation (ρ 0.325→0.645), at the cost of diluting single-node activation (0.833→0.475). All 21 paired tests give p=0.001953—the minimum attainable one-sided p at n=10 (all seeds concordant), which indicates uniform direction and nothing more. We report this as a design-level diagnostic (the fix is an ontology-design decision, and the production graph was not modified), not as behavioral evidence.

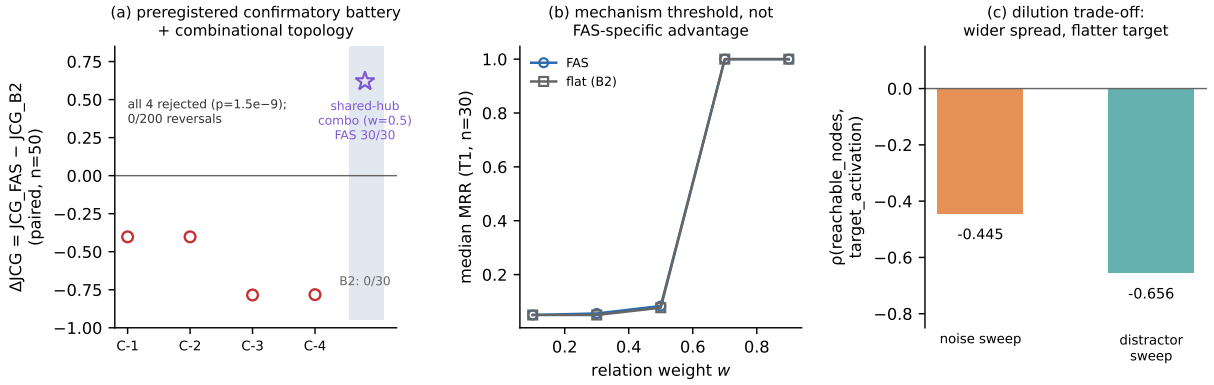


Figure 5: The diffusion applicability boundary. (a) Preregistered confirmatory family: all four ΔJCG negative (Holm p=1.5 × 10^{-9}); the only positive point is the shared-hub topology (star, single-weight w=0.5 design). (b) Weight sweep: both methods reach MRR=1.0 together at w=0.7. (c) Dilution trade-off: the wider the spread, the flatter the target activation (ρ=−0.445, −0.656).

5.4 Determinism re-run of the decision-head family

Because the beacon and routing families use a real LLM decision head (temperature 0), we re-ran a 5-seed × 4-condition subset of the beacon campaign in a fresh output directory and compared against the archived runs. The result is an honest negative: **task outcomes** (success, per condition) match in 20/20 re-runs, but **full action sequences** match bit-for-bit in only 7/20—the remote endpoint is not bit-deterministic even at temperature 0. We therefore claim trajectory-level determinism only for the zero-LLM families (bit-identical by construction and verified by re-runs); the decision-head families are outcome-robust but not trajectory-reproducible, and no claim in this paper rests on their exact trajectories (Supplementary Material S10).

5.5 Comparative behavioral evidence (campaign v2)

The protocol above tests activation against closed-form and embedding retrieval on *mechanism* metrics. To test the same questions on *behavior*, we ran a second campaign (360 runs; design, two mid-course confound repairs, and full data in Supplementary Material S15–S17) in which every condition shares the same scripted experience stream, the same menu, the same LLM decision head (temperature 0), and the same decision budget—only the memory form differs.

Goal resumption under neutral instructions (C). After an open-book practice phase (obtain `oak_planks`) and an unrelated distractor phase, the test prompt gives no goal at all ("you have free choice") and the recipe facts are stripped. FAS resumes and completes the abandoned goal in **9/10** paired seeds; the full-history LLM baseline and the no-memory baseline resume in **0/10** (McNemar p=0.0039; Holm p=0.0117). This advantage is regime-dependent: with a longer experience stream that also contains a failure episode, the full-history baseline matches FAS at 10/10 (below)—the salience advantage holds in the short-history regime. Mechanism attribution: a zero-experience control reproduces the same working set (the recipe-closure prior edge `oak_log`→`oak_planks` re-lights the chain when the raw material is observed), so the effect is *observation-triggered activation over prior relational structure*—relational working-set salience—rather than an intention-lifecycle mechanism.

Goal revision is where the same mechanism costs (E). With an infeasible target (tool-gated ore), the full-history baseline stops repeating the failed gather after **1.0** failures on average and pivots immediately; FAS repeats it **10.5** times and pivots in 7/10 runs (late). A joint measurement (below) gives the direction and magnitude of this cost precisely. The two results are the same coin: state that keeps a goal salient under distraction also keeps it salient when the environment invalidates it.

Consumer repairs (E2/F). We then tested whether the failure and goal states are simply not consumed by the decision context. Wiring the production event-frame's negative-polarity encoding into the campaign assembly (failure→entity edge w=−0.8, parameters identical to the production config) reduced repeated failures from 10.5 to **7.2** (one-sided Wilcoxon p=0.011)—statistically significant, still short of the baseline, so residual inertia is design-level (constant goal text, weak ledger prior). Adding a uniform self-graph goal consumer (`current_goal` → working set; production default off) turned the previously bit-identical ON/OFF behavior into a separation: with the registry cleared, the consumer-on loop re-derives

the abandoned goal in 5/5 seeds and its gathering actions rise from 1.0 to **6.0** per run. The self-model's "behavioral irrelevance" was a missing-consumer artifact, not a property of the representation.

Behavioral convergence (I). On a two-chain convergence task (`oak_fence` requires `oak_planks` and `stick` learned in separate experiences), FAS conditions complete 0–2/10 while the raw-history LLM baseline completes **10/10** (Holm p=0.041). Context budget (k=16, 32) and a structure-enriched serialization (explicit action→outcome pairs) both remain at 0/10 (n=30), locating the failure in the *serialization format*: the graph state drops the temporal narrative that the raw stream preserves (Supplementary Material S17).

Credit horizon (B) and long-run stability (J). A deterministic delay probe on the production attribution pipeline shows action→consequence credit forms in 10/10 seeds up to 80 s and in 0/10 at 160 s—the 90 s attribution window is the architecture's credit horizon (an intentional semantic boundary, Section 7). A 100-cycle stability run (zero-LLM) shows bounded graph growth (+3.7% nodes), clamped activation, and no contamination collapse, but also zero promotion (repetition-starved episodes never reach the support threshold)—long-run consolidation remains untested.

Joint measurement (T). On a single task sequence carrying both mechanisms (a failed infeasible goal followed by a neutral free phase), the event-frame encoding replicates its effect (failed-gather repetitions 12.0→7.8, one-sided Wilcoxon p=0.002), the goal consumer adds no resumption increment where the prior-edge path is already at ceiling (all conditions 10/10) and adds no inertia penalty (6.8 vs 7.8, n.s.)—the two consumers coexist, and the combined working set crowds the failed action out of the top-8 entirely. The behavioral trade-off curve between suppression and salience is therefore measurable, and its first two points favor suppression. (One design flaw—inventory carry-over saturating the first resumption metric—was caught and corrected with a fresh-world rerun; both datasets are archived.)

Together these results revise the paper's claim structure: persistent relational state demonstrably changes behavior in one direction (resumption), demonstrably costs in another (revision), is currently bottlenecked at the serialization interface (convergence), and is credit-limited by an explicit design window (delayed consequence). All negative and reverse-direction results are reported at full strength.

5.6 Additional observations

Five further families are compressed here; full data in Supplementary Material S11–S12.

Long chain and priors. A 14-step chain completes 5/5 seeds at 315 ticks; removing graph prior edges *and* the smelting task prior gives 0/5 (the agent cannot plan out-of-recipe transitions); re-adding only the smelting prior restores the bit-identical 315 ticks; −diffusion is faster (259 ticks). With zero priors, the system never enters a learning regime: 0/25 episodes, the goal system locks onto an unreachable target, and the attribution ledger is completely untouched (0 aggregations) while the graph still grows from perception (+29 nodes)—*scaffolded grounded knowledge acquisition*: experience does not automatically become knowledge.

Hebbian, affect, intentions. Hebbian co-activation strengthening changes exactly 3 whitelisted edges (+0.074 total on the dominant edge; total-sum drift +0.079) over 30 episodes with retrieval ranks bitwise unchanged—mechanism isolation without behavioral effect. Scalar affective modulation moves activation quality monotonically with the gain (49.8 / 52.5 / 57.4 for gains 0.85/1.0/1.3; disabled is bitwise identical to neutral) but moves no action-level metric. The continuous-intention lifecycle forms, competes, and decays from real experience (3–4 intentions per run); the re-fire gate never triggers (field temperature 1.61–2.12 vs cold-field threshold 0.35), and the expression level is never reached (peak 0.46 < threshold 0.78)—formation through decay is validated, expression is not.

Explicit routing (exploratory). On four fully-observable decision-head task classes (520/520 valid runs), FAS context shows one directional positive (competing-targets task: 19/20 vs 15/20 without context, uncorrected p=0.046; after correction over the campaign's 280 tests, $p\approx1$), one efficiency deficit (re-routing task: 4.15 vs 2.3/2.8 decisions, p=0.003/0.044), and structural token costs (2.1–3.7×). An earlier routing campaign was rejected by our own assembly audit (the harness never built edges and swallowed 521/521 exceptions) and is archived as invalid. Under this paper's standard, routing sits with the negative results.

Live observations. Two 25-minute live Minecraft sessions (zero LLM in the action path) are observational only: the first grows the graph by +100 nodes/+365 edges in discovery form; the second adds only +35/+95 with behavior dominated by learned goals; the live causal ledger (89 aggregations / 78 hypotheses / 29 promotions) comes entirely from real behavior. The causal bridge from write-back to behavioral improvement is *not* established in the live window—an honest negative observation.

Closure masking. Across many families, completion-layer metrics are systematically insensitive to internal differences (write-back lineages, re-activation conditions, and modulation conditions all complete identically). On tasks whose closure guarantees reachability, the planner alone suffices and activation-level

knowledge falls outside the decision envelope; evaluations that report only closure-guaranteed completion rates will systematically fail to detect memory phenomena.

6 Discussion

6.1 What FAS demonstrates

Shown, within the exact scope of this protocol: (i) under mechanism-level scaffolding, embodied experience converts into relational structure absent from the initial graph and all injected sources (0 vs 13 nodes; 15 promotions); (ii) that structure persists and re-activates in changed contexts (4/12 vs 0/8 vs 0/0); (iii) two named consumers (ledger, event framework) measurably change behavior (13 vs 16 ticks; 5/5 first-action flips); (iv) under information asymmetry, persistent relational reuse yields a large behavioral difference (20/20 vs 0/20), fully reproducible by activation-free embedding retrieval; (v) the convergence mechanism runs (200/200) yet its incremental value over flat retrieval is preregistration-rejected (0/200 reversals), with the shared-hub exception delimited by the 84-cell sweep. **Not shown:** general intelligence, open-ended learning from scratch, universal superiority over LLM memory systems, causal understanding, universal diffusion benefits, robust routing value, or a behavioral Hebbian benefit. Each item is accompanied by its evidence or its absence.

6.2 Why flat retrieval usually suffices—and when it does not

The negative results organize into a testable conditional statement rather than a slogan. When the task only asks "retrieve the related thing," flat retrieval does the same work at lower complexity (beacon: D-flat 20/20; campaign: ΔJCG<0). The mechanism-level reason is dilution: the wider the diffusion front, the lower the target activation ($\rho=-0.445$, -0.656), which also explains why $-$diffusion is faster on the long chain (259 vs 315 ticks). The sweep sharpens the exception into a region: *activation-based access pays when the task requires multi-relation convergence and the relevant edges are too weak for direct ranking to resolve the target* (all 11 FAS-advantage cells have $w\leq 0.5$, $d\leq 1$); with strong edges, flat ranking resolves everything; with deep weak paths, diffusion dilutes faster than the closed-form product decays (all 9 flat-advantage cells). The guarded hypothesis is: *when decision-relevant information is absent from current observation and the task structure requires multi-relation convergence, activation-based access has positive value over flat retrieval; when information is observable or the task is nearest-neighbor, that value is zero or negative.* Four evidence lines meet at this hypothesis—the beacon (information asymmetry), the shared-hub cells (convergence), the rejected campaign (nearest-neighbor), and the routing tasks (observable information)—none of them stamps it alone.

6.3 The role of scaffolding

The zero-prior result (0/25 with an untouched ledger) shows scaffolding is not a convenience but a precondition: the goal system locks onto an unreachable target before any attributable experience exists, and the learning channel starves. FAS is therefore *scaffolded grounded* rather than *learning from scratch*: the architecture must provide the mechanisms (attribution window, thresholds, mechanics tables) that make experience readable and writable. This holds for any system that must act before it can learn, and it is both a necessary transformation and a boundary of capability.

6.4 Why the unified graph matters

The three functional evidence chains (re-activation carried by episodic structure, exact ledger transfer onto shared candidates, suppression flips through polarity-slot edges) exist because experience, semantics, and action candidates *share one addressable structure*: in separated stores, each coupling would require custom bridging; here they are natural consequences of graph traversal and graph-based activation. The value of the unified graph is of the form *observable consequences*, and the negative results (closure masking, sub-threshold Hebbian drift) delimit what it cannot do: representational changes do not automatically become behavioral changes.

6.5 Design implication: direction semantics

The relation-direction diagnostic (Section 5.3; full data in Supplementary Material S9) generalizes into an ontology-design rule: *bake per-relation-class direction semantics into the ontology rather than assume prop-*

agation will find its way—recipe-class relations forward, observation-co-occurrence relations bidirectional. On the measured recipe graph (44 nodes/53 edges), forward two-hop reachability is structurally zero because every relation target is a sink; the improvement after adding reverse projections is a design-level verification, not an emergent property.

7 Limitations

(1) All experiments run in a deterministic sandbox with synthetic topologies; paired seeds instantiate different random instances, not real-world variance, and all significance statements quantify directional consistency, not population effects. (2) The beacon's 0/20 baselines are existence verifications built into the task design (closed-book gating), not independent statistical findings. (3) Single environment family, single Minecraft-style domain; no third-party benchmarks; no external-environment validation (e.g., a Crafter-style equivalent) was performed. (4) Subject experiments use five deterministic seeds—descriptive statistics only; paired experiments (n=10, n=20, n=50) carry the directional claims with stated corrections. (5) Semantic write-back structure is not consumed by the planner; learned knowledge affects behavior through the ledger, event-framework, and re-activation channels only. (6) Zero-prior conditions test locking, not learning ability. (7) Not implemented: prospective memory, full perception pipeline, symbolic reasoning, RL policy nodes; several implemented mechanisms are not experimentally validated here. (8) Event-framework behavioral results come from experimental assembly of a production-default-off mechanism. (9) Scaffolding dependence is not quantified. (10) Experimental graphs are small (44–1000 nodes); scale limits all mechanism conclusions. (11) Routing tasks are fully observable at decision time and the negative results are conditional on that; token costs and directional differences are exploratory. (12) The mechanism campaign and relation-direction diagnosis are mechanism-level evidence without production behavioral tests; the shared-hub sweep, while preregistered-style and frozen before execution, is still confined to synthetic hub topologies with a closed-form flat surrogate, not an embedding retriever. (13) The beacon is a single task family (craft domain, closed book); cross-task generality of the information-asymmetry benefit is untested. (14) The decision-head families depend on a remote LLM endpoint at temperature 0; bit-level determinism is verified only by re-running (Section 5.4) (15) The attribution window (90 s) is a hard credit horizon: delayed consequences beyond it are never assigned (Section 5.5). (16) The decision-head context serialization drops the temporal narrative of the experience stream; behavioral convergence tasks therefore fail at this interface. (17) The resumption advantage over full-history baselines is regime-dependent (short-history regime only).

8 Conclusion

On a unified cognitive graph, with mechanism-level scaffolding, embodied experience becomes new, typed, evidence-weighted relational knowledge: it persists, re-activates across contexts, and reaches behavior through named consumer channels in measurable, deterministic ways. On the information-asymmetric beacon task, persistent relational reuse yields a 20/20 vs 0/20 behavioral difference, fully reproduced without activation. Spreading activation is a real, measurable access mechanism whose incremental value over flat retrieval is preregistration-rejected everywhere except a now-parameterized shared-hub region: weak edges, shallow convergence paths. The boundaries are equally clear: no-scaffolding learning regimes are never entered; closure-guaranteed tasks mask internal differences; diffusion's footprint is knowledge access, not action selection. We offer the architecture and the protocol—deterministic, zero-LLM, fully archived, with two reported assembly audits and a preregistered rejection—as a substrate for studying when experience becomes knowledge that changes behavior, and we hand over all negative results as part of the map.

Data and code availability. All experiments are deterministic and fully archived (raw run outputs, graph snapshots, statistics files); a one-click reproduction script, the preregistration records, and the claim-to-evidence mapping ship with the artifacts (Supplementary Material S14).

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